STUDYSMART

A Learning-Based Android Application

Offline-first academic study and revision

Tribhuvan University
Xavier International College

Bachelor of Computer Applications (BCA)
8th Semester

Final Project Presentation

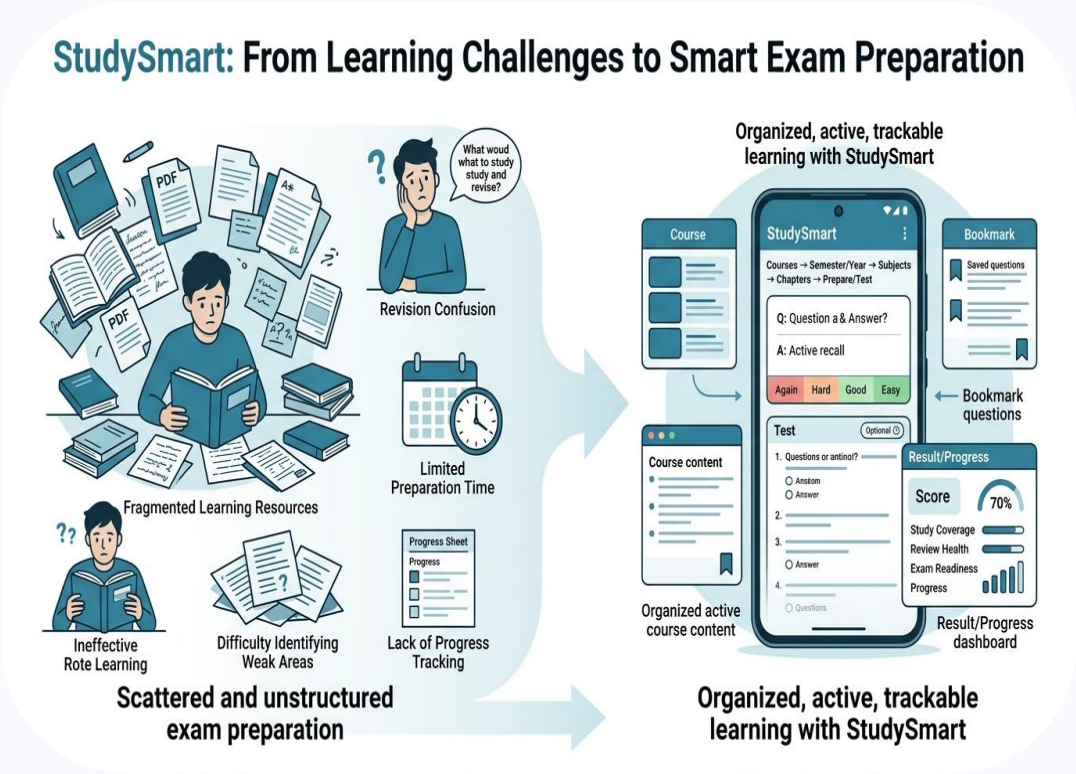
Prepared By:

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Introduction

StudySmart is designed to turn scattered exam preparation into a structured learning workflow.

- Organizes academic content by course, academic period, subject, chapter and question.
- Supports active recall using question–answer preparation cards.
- Records learning ratings and schedules due review.
- Provides written mock tests with transparent keyword-based evaluation.
- Keeps the Core Alpha learning loop available offline.



OFFLINE-FIRST

ANDROID

ACTIVE RECALL

Problem Statement

Students may have learning materials — but still lack a consistent process for revision.

Scattered materials

Questions and answers are often spread across notes, PDFs, websites and separate applications.

Weak review tracking

Students may not know which questions were difficult or when they should revisit them.

Manual self-testing

Creating and checking written mock tests can require repetitive manual effort.

Limited evidence

Study history, written answers and test evidence may not be stored consistently.

StudySmart connects browsing → preparation → review → written testing → results in one focused local workflow.

Objectives

General Objective

Develop an offline-first Android application for structured study, due review, written mock testing and inspectable results.

1. **Structured academic browsing**
   Course → period → subject → chapter → question
2. **Preparation & review**
   Answer reveal with Again / Hard / Good / Easy ratings
3. **Deterministic scheduling**
   Convert ratings into explicit due-review intervals
4. **Written mock tests**
   Create fixed chapter-scoped attempts and persist answers
5. **Transparent evaluation**
   Score answers using predefined weighted keywords
6. **Verification**
   Test domain rules, persistence and critical navigation

Literature Review

StudySmart is informed by established study concepts, but its Core Alpha uses deliberately simple and explainable rules.

Retrieval Practice

Learners actively recall an answer before revealing the reference answer. StudySmart applies this through question–answer preparation cards.

Spaced / Distributed Review

Study items return after intervals rather than being treated equally. The project uses deterministic intervals instead of claiming an optimized scientific scheduler.

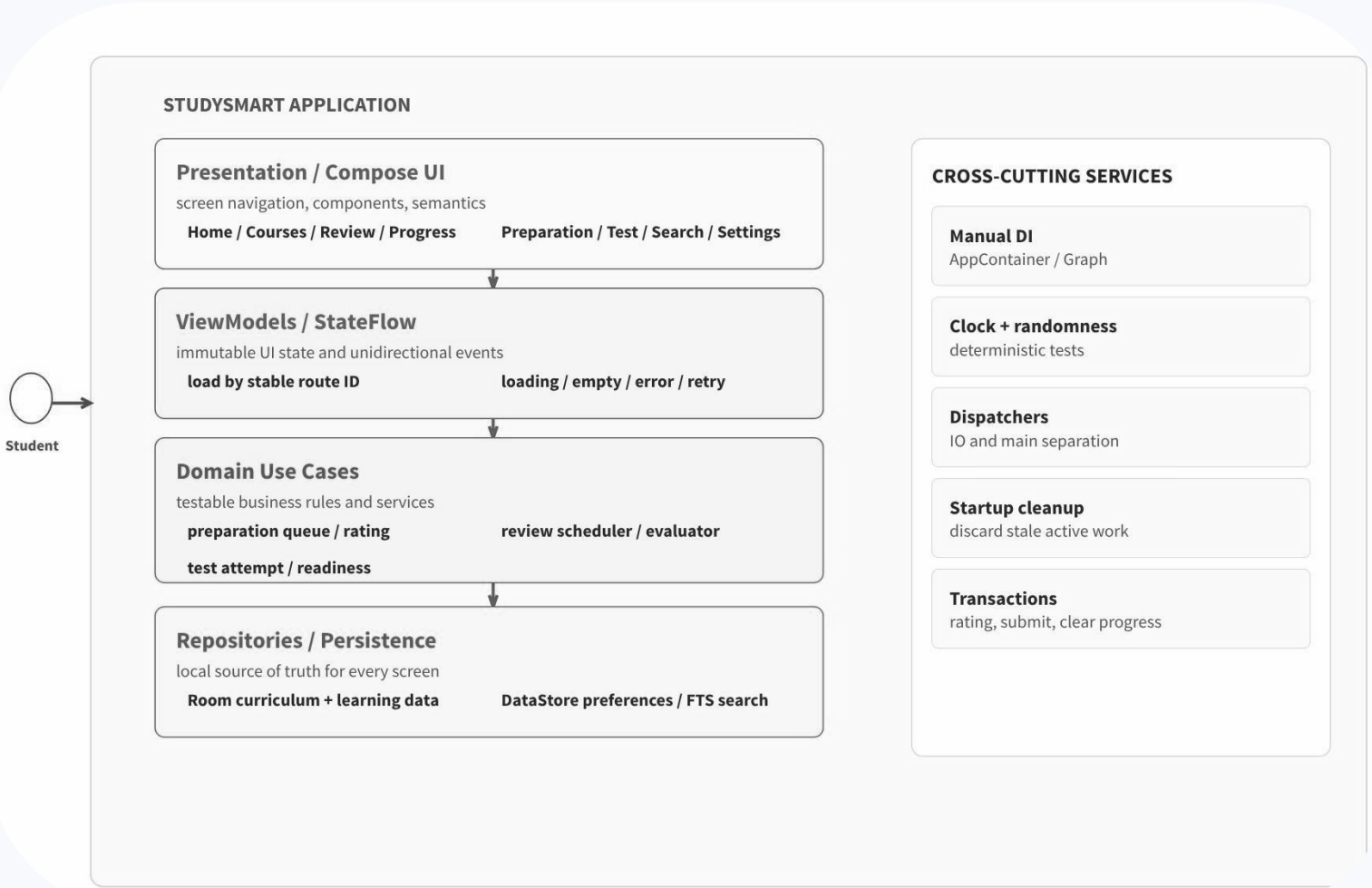
Mobile & Offline Learning

A phone-based local workflow keeps the core study loop available without requiring network access during browsing, preparation, review or testing.

Positioning

The contribution is the integration of these concepts into a small, explainable BCA project rather than a claim of superiority over large learning platforms.

System Design



Core design decisions

- Offline-first local data source
- Course → Academic Period → Subject → Chapter → optional Topic → Question
- Room for relational study data
- DataStore for small preferences
- Repository + View Model separation
- Testable domain services for business rules

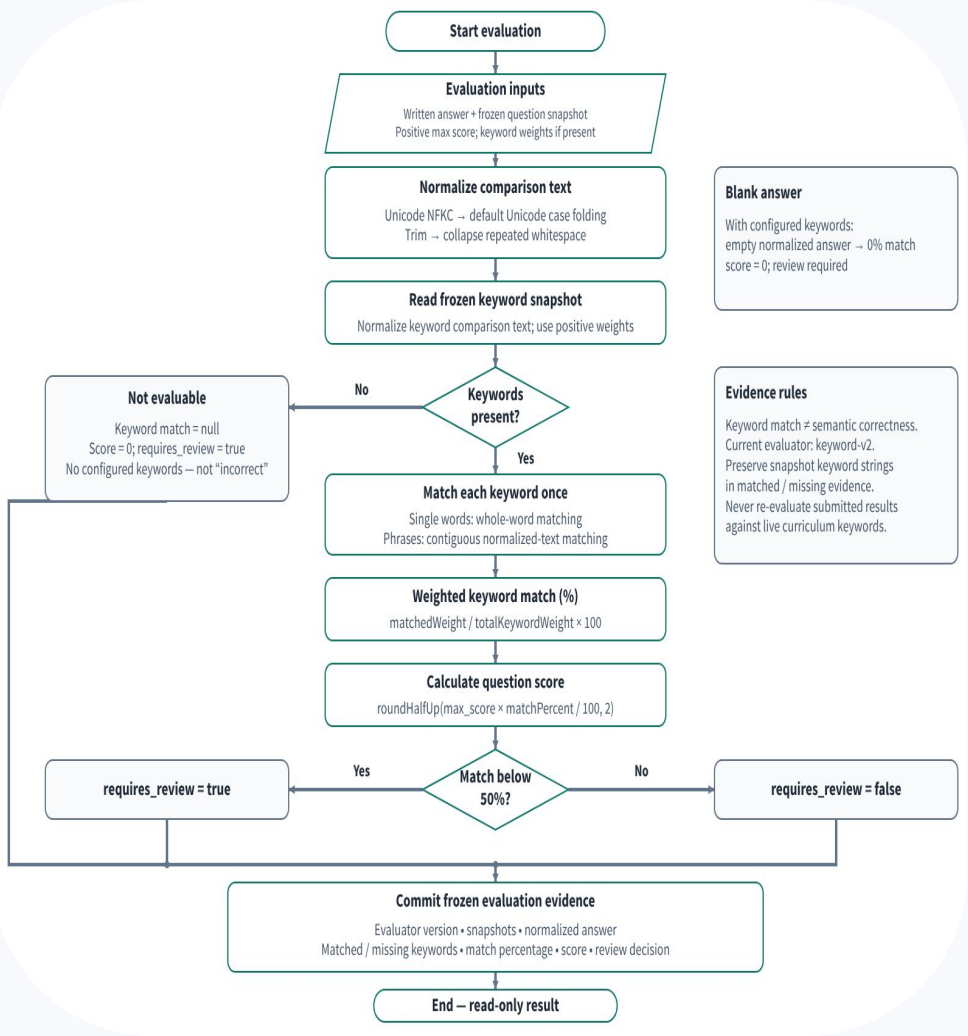
Core Algorithms

Review scheduling

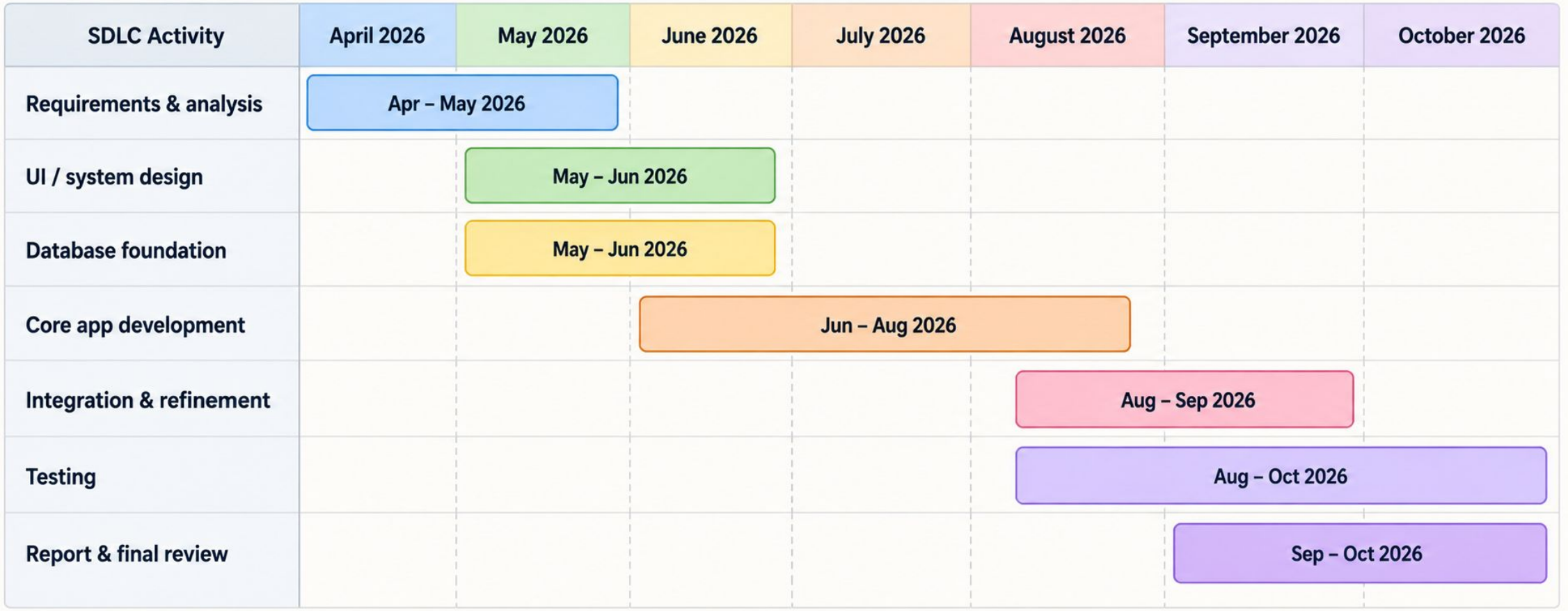
Again	Hard	Good	Easy
10 minutes	1 day	3 days	7 days

Keyword answer evaluation

- Normalize the written response.
- Match predefined keywords.
- Apply keyword weights.
- Calculate a bounded score.
- Store result evidence for the submitted attempt.



Gantt Chart


SDLC Activity
April 2026
May 2026
June 2026
July 2026
August 2026
September 2026
October 2026
Requirements & analysis
Apr – May 2026
UI / system design
May – Jun 2026
Database foundation
May – Jun 2026
Core app development
Jun – Aug 2026
Integration & refinement
Aug – Sep 2026
Testing
Aug – Oct 2026
Report & final review
Sep – Oct 2026

Implementation

Technology stack

Kotlin
Application language

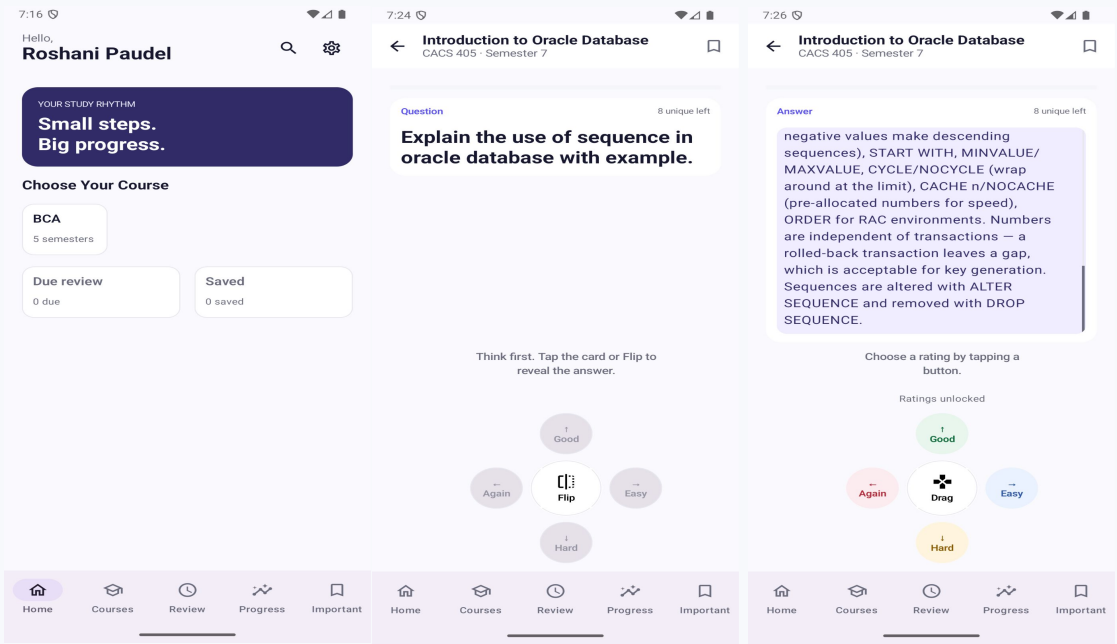
Jetpack Compose
Declarative Android UI

Material 3
Design system

Room
Relational local storage

DataStore
Preferences

Coroutines / Flow
Asynchronous & reactive state



Testing

Testing focuses on the rules and workflows that make the learning loop reliable.

Domain / Unit	Review scheduler, queue rules, keyword evaluator
Persistence	Room constraints, repositories, transactions, snapshots
ViewModel	Screen state, retry/error behavior, lifecycle state
Navigation / UI	Critical Compose navigation and user flows

Test Execution Screenshot

```
$ ./gradlew :app:testDebugUnitTest --rerun-tasks
> Task :app:compileDebugUnitTestKotlin
> Task :app:compileDebugUnitTestJava NO-SOURCE
> Task :app:testDebugUnitTest
BUILD SUCCESSFUL in 1m 38s
381 actionable tasks: 381 executed
Test run finished after 95432 ms
[      381 tests found
[        0 tests skipped
[      381 tests started
[        0 tests failed
[      381 tests successful
```

Automated Test Summary

381 Passed	0 Failed	0 Skipped	381 Total Executed (JVM Unit Tests)

- Command: `./gradlew :app:testDebugUnitTest --rerun-tasks`
- Execution date: 6 October 2026
- Results based on fresh JUnit XML output
- Covers major domain rules and selected state/repository transformation behaviour

Representative Unit Test Cases

Test Class	No. of Test Cases	Result
AcademicPeriodTypeTest	2	Passed
PreparationSelectionEngineTest	12	Passed
PreparationQueueEngineTest	13	Passed
ReviewSchedulerTest	6	Passed
KeywordAnswerEvaluatorTest	9	Passed
ReadinessCalculatorTest	7	Passed
CurriculumRepositoryTest (transformation case)	1	Passed

Integration Test Status

Test Case	Status
IT-01 Curriculum DAO/repository output	Partially tested
IT-02 Course selection/persisted context	Partially tested
IT-03 Preparation session persistence	Implemented (manual verification required)
IT-04 Rating history + scheduling transaction	Implemented (manual verification required)
IT-05 Due-review session creation	Partially tested
IT-06 Fixed test attempt creation	Partially tested
IT-07 Draft persistence/navigation	Partially tested
IT-08 Submission/evaluation/result commit	Partially tested
IT-09 Weak-question retest	Implemented (manual verification required)
IT-10 Startup cleanup	Implemented (manual verification required)

Other Testing Categories

Category	Status	Details
Instrumentation / UI Tests	Not executed	54 androidTest files (~286 test annortations) Inventory only
Functional / System Tests	Not executed	FT-01 – FT-23 Manual verification required
User Acceptance Testing (UAT)	Not conducted	No participant testing evidenced
Performance Testing	Not conducted	No formal benchmarking performed

Conclusion

StudySmart demonstrates a focused learning workflow that is small enough to explain, test and extend.

Structured

Academic browsing follows a consistent hierarchy rather than scattered files.

Active

Preparation requires recall before answer reveal and captures learning ratings.

Repeatable

Due-review rules are deterministic and inspectable.

Measurable

Written mock tests preserve answers and transparent keyword evidence.

Core contribution: one offline-first Android workflow connecting study content, preparation, review and written self-assessment.

Future Improvements

Accounts & sync

Authentication, multiple users, cloud backup and synchronization

Teacher / admin tools

Content management, monitoring and controlled publishing

Smart reminders

Notifications, review reminders and daily study goals

More question types

MCQ, true/false and richer assessment formats

Advanced evaluation

Carefully validated semantic or AI-assisted grading

Cross-platform access

Web or iOS delivery while preserving the learning model

Future work should extend the verified Core Alpha rather than hide its current scope limitations.

THANK YOU

Questions & Discussion

StudySmart

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